unambiguous, completely identifiable time and place where this happens: It is marked in Figure 6.6.

How does this description change if, when we closely examine the objects we thought were zero-dimensional points, they turn out to be one-dimensional strings? The basic process of interaction is the same, but now the objects on a collision course are oscillating loops, as shown in Figure 6.7. If these loops are vibrating in just the right resonance patterns, they will correspond to an electron and a positron on collision course, just as in Figure 6.6. Only when examined at the most minute distance scales, far smaller than anything our present technology can access, is their true stringlike character apparent. As in the point-particle case, the two strings collide and again annihilate each other in a flash. The flash, a photon, is itself a string in a particular vibrational pattern. Thus, the two incoming strings interact by merging together and producing a third string, as shown in Figure 6.7. Just as in our point-particle description, this string travels a bit, and then releases the energy derived from the two initial strings by dissociating into two strings that travel onward. Again, from any but the most microscopic perspective, this will look just like the point-particle interaction of Figure 6.6.

There is, however, a crucial difference between the two descriptions. We emphasized that the point-particle interaction occurs at an identifiable point in space and time, a location that all observers can agree on. As we shall now see, this is not true for interactions between strings. We will show this by comparing how George and Gracie, two observers in relative motion as in Chapter 2, would describe the interaction. We will see that they do not agree on where and when the two strings touch for the first time.

To do so, imagine that we view the interaction between two strings with a camera whose shutter is kept open so that the whole history of the process is captured on one piece of film.10 We show the result—known as a string world-sheet—in Figure 6.7(c). By “slicing” the world-sheet into parallel pieces—much as one slices a loaf of bread—the moment-by-moment history of the string interaction can be recovered. We show an example of this slicing in Figure 6.8. Specifically, in Figure 6.8(a) we show George, intently focused on the two incoming strings, together with an attached plane that slices